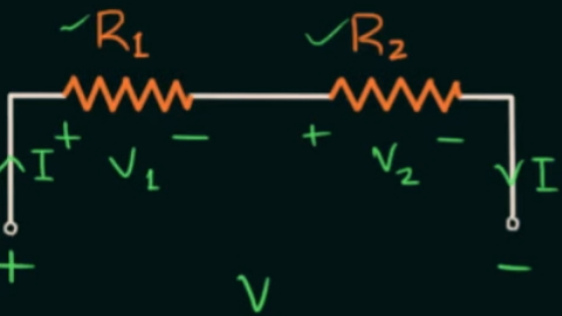


series $\rightarrow I \rightarrow$ same



$$R_{eq} = R_1 + R_2$$

$$I = \frac{V}{R_{eq}} = \frac{V}{R_1 + R_2}$$

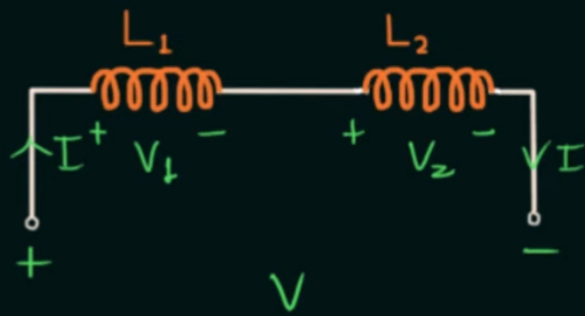
$$V_1 = IR_1 = \frac{V}{R_1 + R_2} \cdot R_1$$

total in.
 own
 del.

$$V_2 = IR_2 = \frac{V}{R_1 + R_2} \cdot R_2$$

$$V_n = \frac{V}{R_1 + R_2 + \dots + R_n} \cdot R_n$$

Voltage Divider Rule

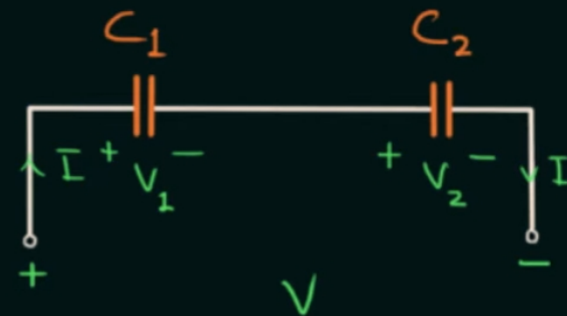


$$V = L_{eq} \frac{dI}{dt} \Rightarrow \frac{dI}{dt} = \frac{V}{L_1 + L_2}$$

$$V_1 = L_1 \frac{dI}{dt} = \frac{V}{L_1 + L_2} \cdot L_1$$

$$V_2 = L_2 \frac{dI}{dt} = \frac{V}{L_1 + L_2} \cdot L_2$$

$$V_n = \frac{V}{L_1 + L_2 + \dots + L_n} \cdot L_n$$



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

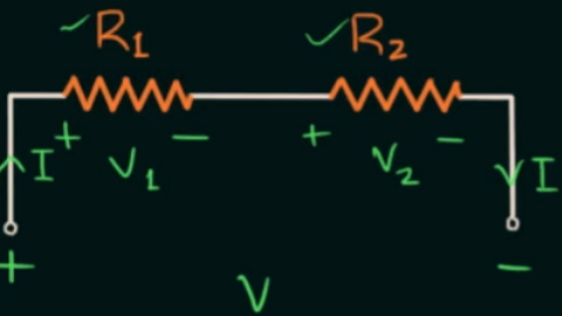
$$Q = C_{eq} V = \frac{C_1 C_2}{C_1 + C_2} \cdot V$$

$$V_1 = \frac{Q}{C_1} = \frac{C_1 C_2 V}{C_1 + C_2} \cdot \frac{1}{C_1} = \frac{V}{C_1 + C_2} \cdot C_2$$

$$V_2 = \frac{Q}{C_2} = \frac{V}{C_1 + C_2} \cdot C_1$$

How $V_3 = ?$

series $\rightarrow I \rightarrow$ same



$$R_{eq} = R_1 + R_2$$

$$I = \frac{V}{R_{eq}} = \frac{V}{R_1 + R_2}$$

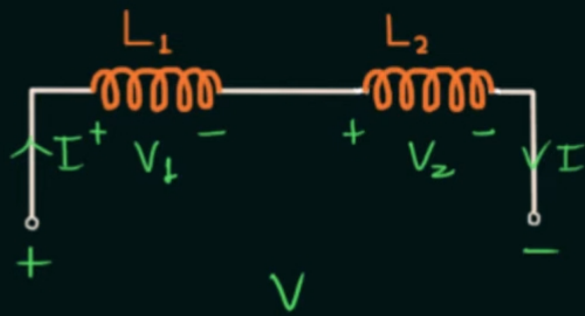
$$V_1 = IR_1 = \frac{V}{R_1 + R_2} \cdot R_1$$

total v. $\swarrow$
own $\nwarrow$
res.

$$V_2 = IR_2 = \frac{V}{R_1 + R_2} \cdot R_2$$

$$V_n = \frac{V}{R_1 + R_2 + \dots + R_n} \cdot R_n$$

Voltage Divider Rule

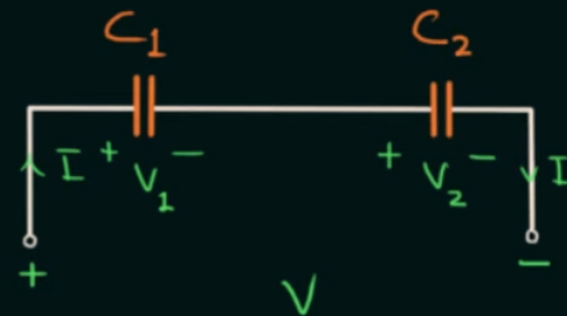


$$V = L_{eq} \frac{dI}{dt} \Rightarrow \frac{dI}{dt} = \frac{V}{L_1 + L_2}$$

$$V_1 = L_1 \frac{dI}{dt} = \frac{V}{L_1 + L_2} \cdot L_1$$

$$V_2 = L_2 \frac{dI}{dt} = \frac{V}{L_1 + L_2} \cdot L_2$$

$$V_n = \frac{V}{L_1 + L_2 + \dots + L_n} \cdot L_n$$



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$Q = C_{eq} V = \frac{C_1 C_2}{C_1 + C_2} \cdot V$$

$$V_1 = \frac{Q}{C_1} = \frac{C_1 C_2 V}{C_1 + C_2} \cdot \frac{1}{C_1} = \frac{V}{C_1 + C_2} \cdot C_2$$

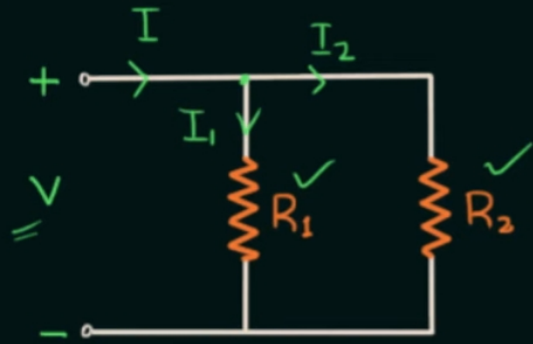
$$V_2 = \frac{Q}{C_2} = \frac{V}{C_1 + C_2} \cdot C_1$$

How $V_3 = V_{Total} \cdot \frac{C_{eq}}{C_3}$

Do read this for Information on How to calculate Voltage through Capacitors with more than 2 capacitors :

<https://gemini.google.com/share/3ffa28f629a5>

parallel $\rightarrow V \rightarrow$ same



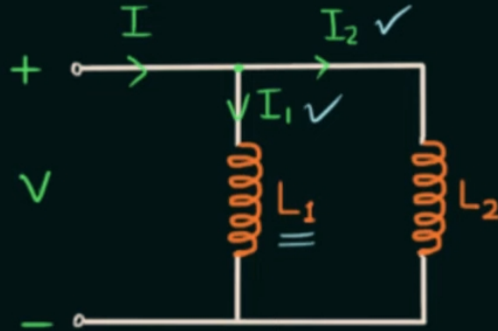
$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$V = I R_{eq} \Rightarrow V = \frac{I R_1 R_2}{R_1 + R_2}$$

$$I_1 = \frac{V}{R_1} = \frac{I R_1 R_2}{(R_1 + R_2) R_1} = \frac{I}{R_1 + R_2} \cdot R_2$$

$$I_2 = \frac{V}{R_2} = \frac{I}{R_1 + R_2} \cdot R_1$$

✓ Current Divider Rule



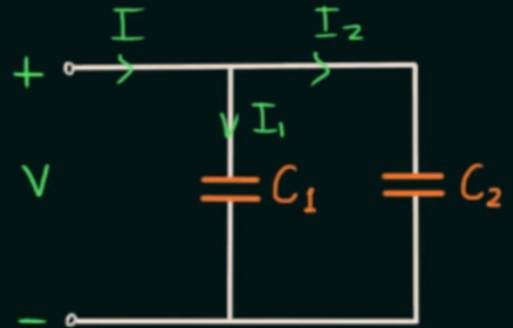
$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

$$I = \frac{1}{L_{eq}} \int V dt \quad \text{and} \quad V = L_{eq} \frac{dI}{dt}$$

$$\int V dt = I L_{eq} = \frac{I L_1 L_2}{L_1 + L_2}$$

$$I_1 = \frac{1}{L_1} \int V dt = \frac{I}{L_1 + L_2} \cdot L_2$$

$$I_2 = \frac{1}{L_2} \int V dt = \frac{I}{L_1 + L_2} \cdot L_1$$



$$C_{eq} = C_1 + C_2$$

$$I = C_{eq} \frac{dV}{dt} \Rightarrow \frac{dV}{dt} = \frac{I}{C_1 + C_2}$$

$$I_1 = C_1 \frac{dV}{dt} = \frac{I}{C_1 + C_2} \cdot C_1$$

$$I_2 = C_2 \frac{dV}{dt} = \frac{I}{C_1 + C_2} \cdot C_2$$

**Do read this for Information for more than 2
R, L, C :-**

<https://gemini.google.com/share/fc78521131cf>